

MATHEMATICS 2270

Quiz 6, Fall semester 2012

Name: KEY

1. (11 pts) Let $A = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}$.

a) Find an orthonormal basis of eigenvectors for A .

b) Compute A^n and $\exp(At)$.

c) Let $x = x(t)$ and $y = y(t)$ be functions of t . Solve

$$\frac{d}{dt}(x, y)^T = A(x, y)^T \quad (x(0), y(0))^T = (2, -1)^T.$$

$$\begin{cases} \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \\ x(0) = 2 \\ y(0) = -1 \end{cases}$$

$$\begin{vmatrix} 2-\lambda & 3 \\ 3 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 9 = \lambda^2 - 4\lambda - 5 = (\lambda-5)(\lambda+1)$$

$$\lambda = 5 \quad N \begin{pmatrix} -3 & 3 \\ 3 & -3 \end{pmatrix} = \langle \begin{pmatrix} 1 \\ 1 \end{pmatrix} \rangle \quad \lambda = -1 \quad N \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix} = \langle \begin{pmatrix} 1 \\ -1 \end{pmatrix} \rangle$$

$$\text{ON basis : } \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} \quad \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} \quad S = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \quad S^T A S = \begin{pmatrix} 5 & 0 \\ 0 & -1 \end{pmatrix} = \Lambda$$

$$S^{-1} = S^T$$

$$A^n = S \Lambda S^T \quad \underline{A^n} = S \Lambda^n S^T = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 5^n & 0 \\ 0 & (-1)^n \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} =$$

$$\begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 5^n/\sqrt{2} & 5^n/\sqrt{2} \\ (-1)^n/\sqrt{2} & -(-1)^n/\sqrt{2} \end{pmatrix} = \begin{pmatrix} \frac{5^n + (-1)^n}{2} & \frac{5^n - (-1)^n}{2} \\ \frac{5^n - (-1)^n}{2} & \frac{5^n + (-1)^n}{2} \end{pmatrix}$$

$$\underline{e^{At}} = S e^{\Lambda t} S^T = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} e^{5t} & 0 \\ 0 & e^{-t} \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} \frac{e^{5t}}{\sqrt{2}} & \frac{e^{5t}}{\sqrt{2}} \\ \frac{e^{-t}}{\sqrt{2}} & -\frac{e^{-t}}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{e^{5t} + e^{-t}}{2} & \frac{e^{5t} - e^{-t}}{2} \\ \frac{e^{5t} - e^{-t}}{2} & \frac{e^{5t} + e^{-t}}{2} \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = e^{At} \cdot \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{e^{5t}}{2} + \frac{3e^{-t}}{2} \\ \frac{e^{5t}}{2} - \frac{3e^{-t}}{2} \end{pmatrix}$$